

Impact of Lifestyle Factors on College Students' GPA

A STATISTICAL ANALYSIS PROJECT

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Research Question and Importance

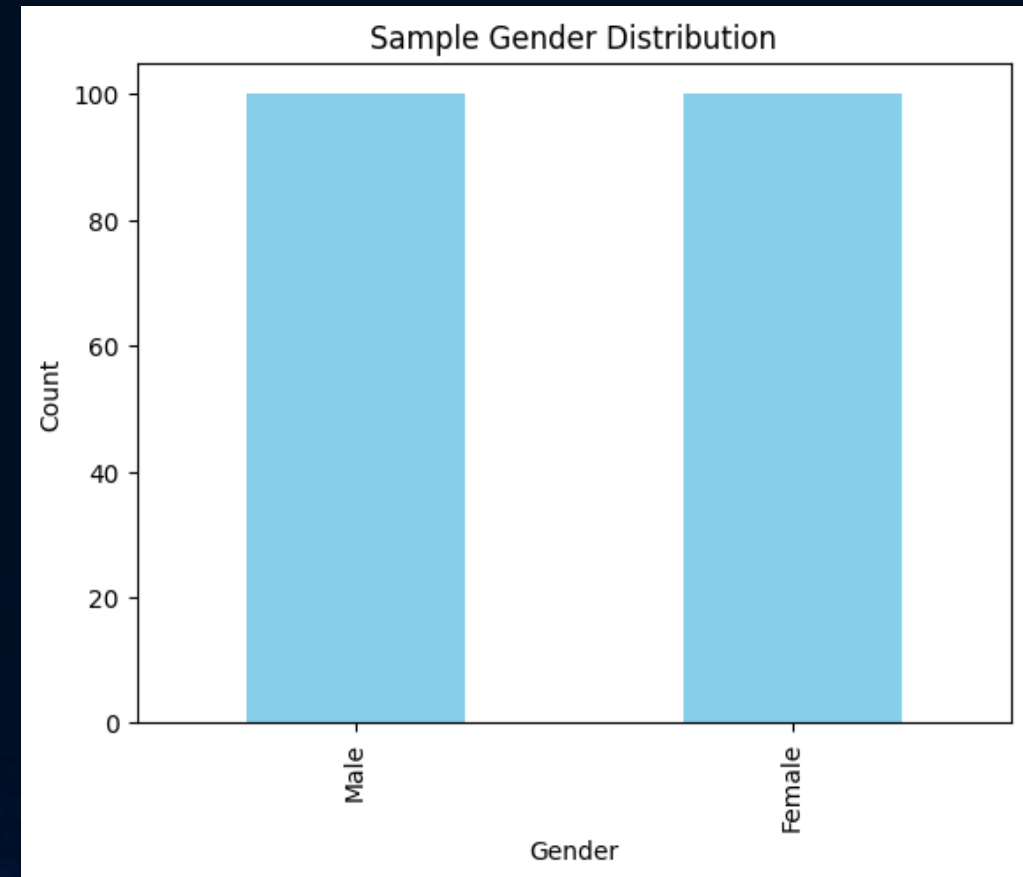
Investigating the impact of Lifestyle Habits on GPA

- How do lifestyle factors like sleep, study hours, exercise, and attendance impact college students' GPA?
- **Why it matters:** Better habits = Better performance
- Understanding these factors can help students optimize their habits for better academic outcomes.

The Population and Sample

Study Design and Sampling Method

- **Population:** All full-time college undergraduate students
- **Sample:** 200 students across different majors
- **Sampling method:** Stratified random sampling (by gender and major)
- **Data Generation:** Synthetic data simulating relationships between lifestyle factors and GPA
- **Bias:** Simulation assumptions (avoids real-world bias, but relies on reasonable assumptions)



Dataset Overview

Variables

- **Dependent Variable (Outcome):**
 - **GPA** – Academic performance measured on a 4.0 scale
- **Independent Variables (Predictors):**
 - **Study Hours** – Average hours per week spent studying
 - **Sleep Hours** – Average hours of sleep per night
 - **Exercise Hours** – Average hours of exercise per week
 - **Attendance Rate** – Class attendance percentage
- **Exploratory Variable:**
 - **Gender** – Categorical variable (Male/Female)

Data Preprocessing Overview

- Data preprocessing is a crucial step in ensuring data quality before analysis. Below are the key steps:
 - Handling missing values
 - Outlier detection and handling
 - Data scaling
 - Categorical encoding (e.g., gender)

Handling Missing Data

- **Step 1:** Check for missing values

```
# Check for missing values  
print(students_data.isnull().sum())
```

- **Step 2:** Drop or impute missing values

```
# Drop missing values if found  
students_data.dropna(inplace=True)
```

Outlier Detection and Handling

- **Method:** IQR (Interquartile Range) for outlier detection.
- **Step 1:** Identify outliers.
- **Step 2:** Remove outliers.

Code:

```
# Detect and remove outliers in 'Study_Hours' using IQR
Q1 = students_data['Study_Hours'].quantile(0.25)
Q3 = students_data['Study_Hours'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
students_data = students_data[(students_data['Study_Hours'] >= lower_bound) & (students_data['Study_Hours'] <= upper_bound)]
```

Data Scaling and Categorical Encoding

- **Scaling:** Standardizing numerical variables (Study Hours, Sleep, Exercise, Attendance).
- **Encoding:** Converting categorical data (e.g., Gender) into binary form.
- **Code for Standardizing numerical variables:**

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaled_cols = ['Study_Hours', 'Sleep_Hours', 'Exercise_Hours', 'Attendance']
students_data[scaled_cols] = scaler.fit_transform(students_data[scaled_cols])
```

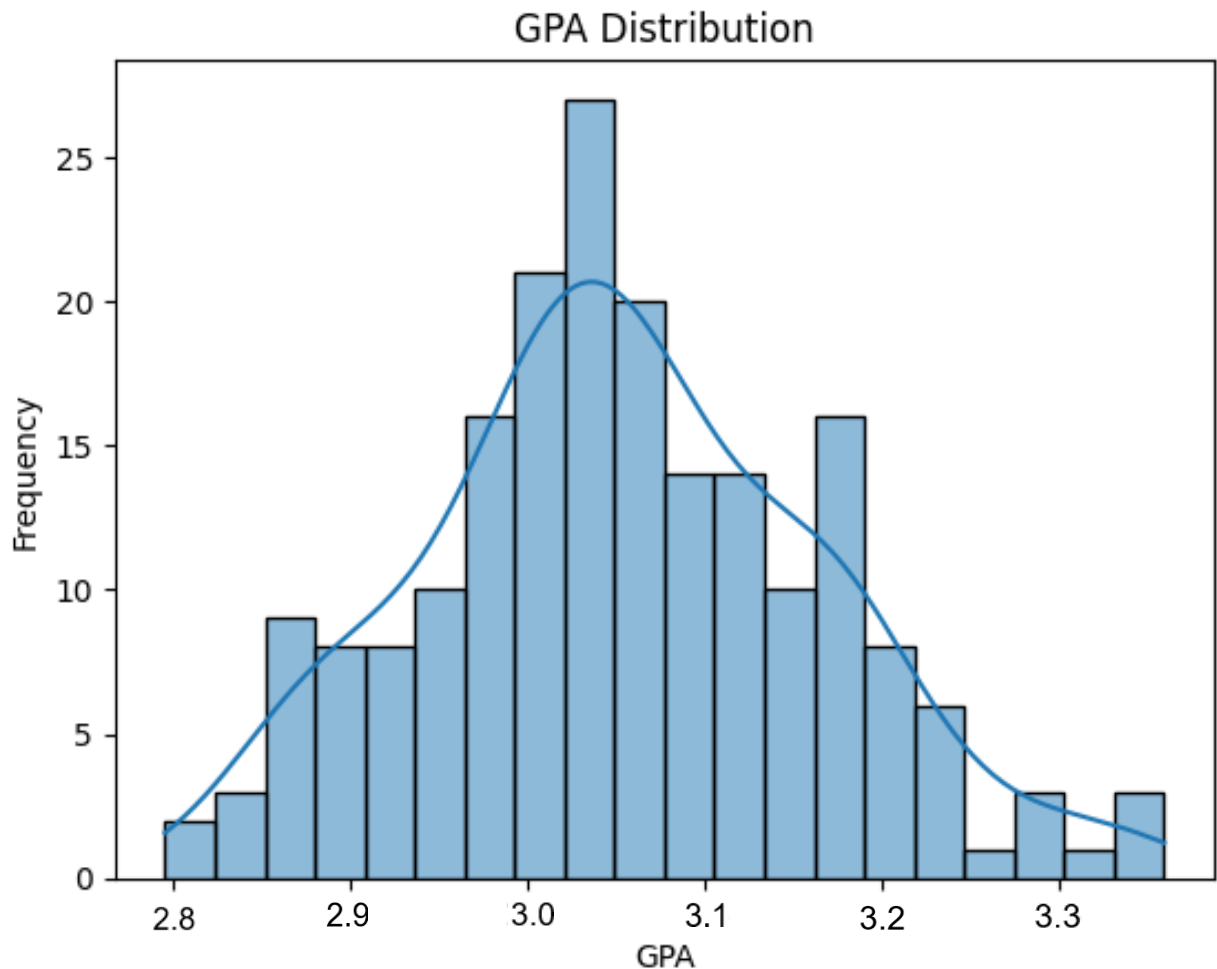
Parameters vs. Statistics

Estimating the Unknown

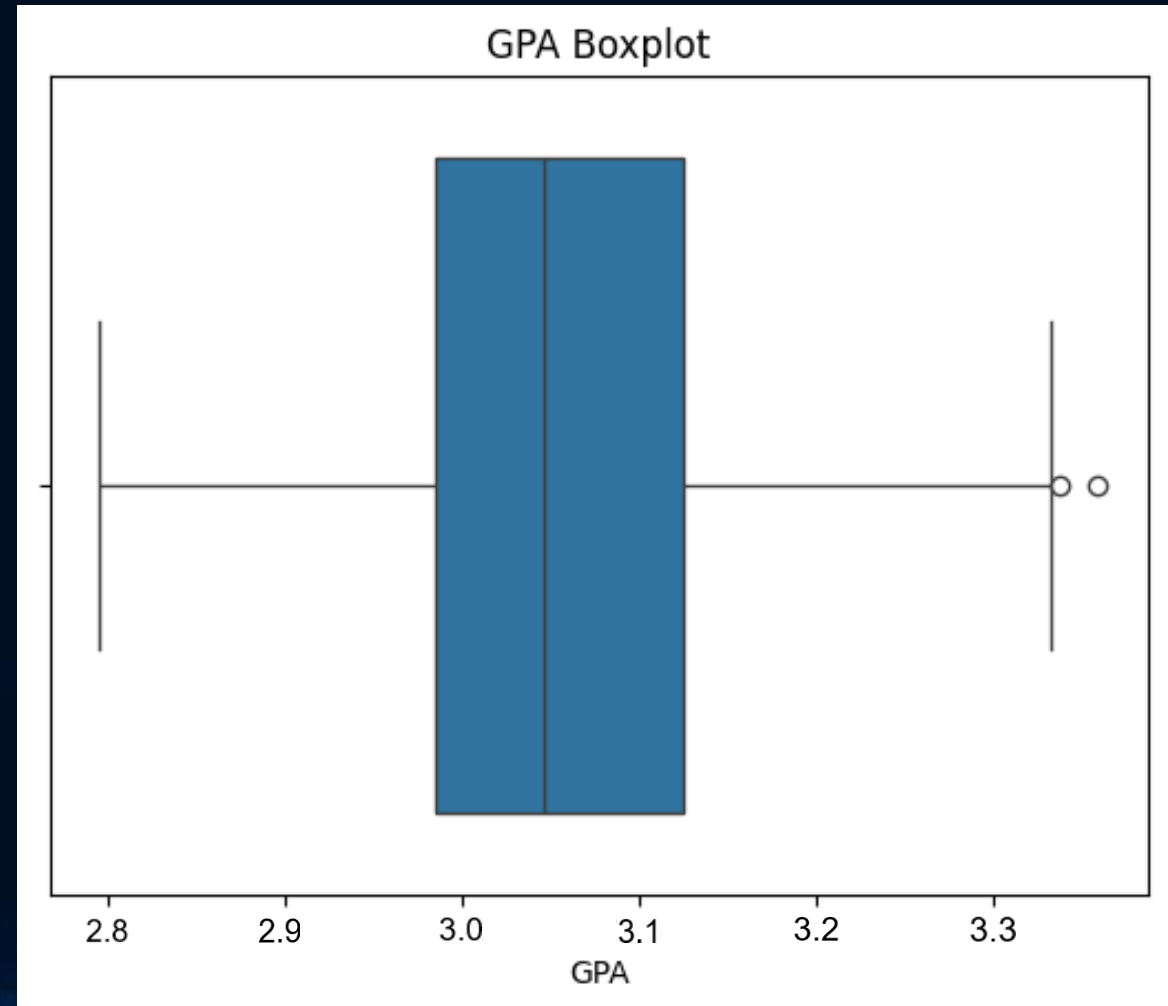
- Parameter = true population GPA (unknown)
- Sample Statistic ($\bar{x}$) = estimated mean GPA = 3.05  **Average GPA of the sample**
- Sample Standard Deviation (s) = 0.41  **Spread of GPA across students**
Tells us how spread out the GPAs are around the average. A smaller value means the GPAs are more consistent
- Sample Size (n): 200

Sample Statistics: GPA Overview

Histogram of GPA (shows distribution)



Boxplot for GPA (shows spread)



Standard Error and 95% Confidence Interval

- Standard Error of Mean (SEM) = $s / \sqrt{n} = 0.41 / \sqrt{200} = 0.029$
- 95% Confidence Interval (CI) = $3.05 \pm 1.96 \times 0.029 = (2.99, 3.11)$

What does that tell us?

SEM shows how accurate our sample mean is as an estimate of the true population mean. A smaller SEM means more precision. The confidence interval tells us we're 95% confident that the true mean GPA lies between 2.99 and 3.11.

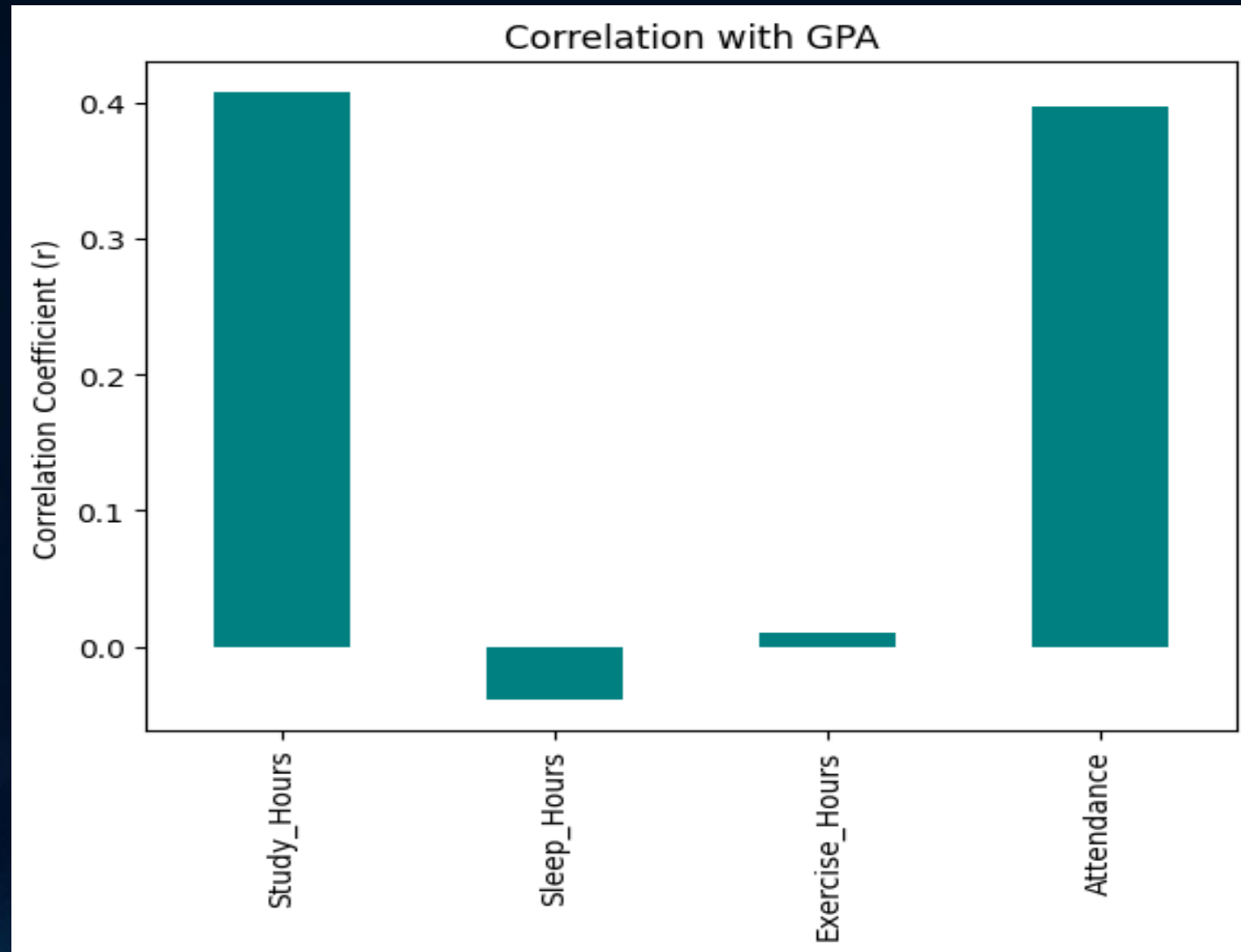
Correlation with GPA

What Matters Most?

- Study Hours: $p = 0.0000$ (Significant relationship with GPA)
- Attendance: $p = 0.0000$ (Significant relationship with GPA)
- Sleep Hours: $p = 0.5798$ (No significant relationship with GPA)
- Exercise Hours: $p = 0.8834$ (No significant relationship with GPA)

Variable	r-value	p-value
Study Hours	0.42	< 0.0001
Attendance	0.41	< 0.0001
Sleep Hours	0.06	0.5798
Exercise Hours	0.009	0.8835

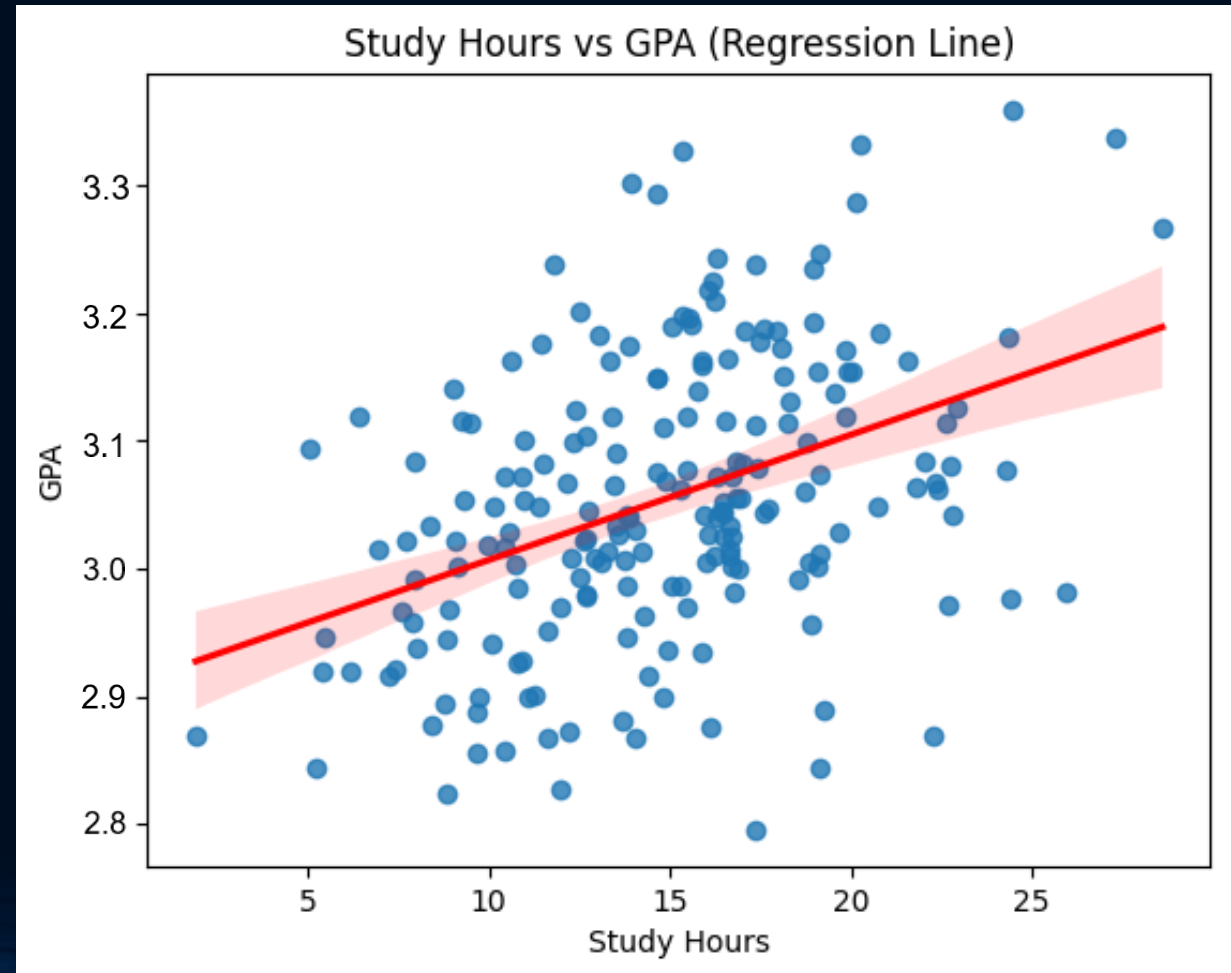
Correlations: What Matters Most?



Linear Regression Summary

Predicting GPA

- Dependent Variable: GPA
- Top Predictor: Study Hours
- Other Factors: Attendance also contributes
- $R^2 = 0.35$ (Explains 35% of GPA variance)



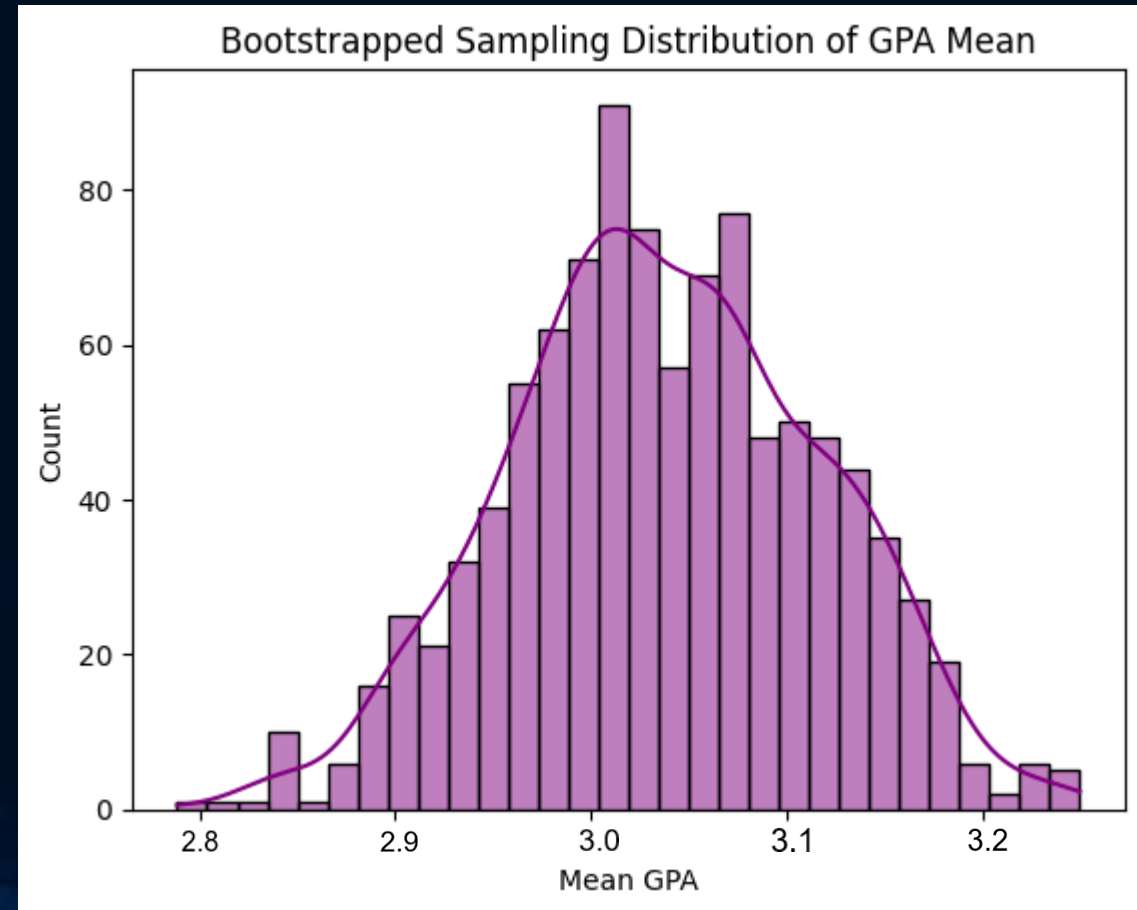
Bootstrapping and Central Limit Theorem (CLT)

Bootstrapping: Estimating Variability When Assumptions Break

- **CLT:** Sampling distributions ~ normal for large n
- **Bootstrap:** Resample data to estimate statistics. Useful when data isn't normal or sample size is small.

What does that tell us?

The Central Limit Theorem lets us make reliable inferences from sample data. Bootstrapping gives us a way to estimate uncertainty without needing normality or large samples.



Hypothesis Testing Example

Does Attendance Matter?

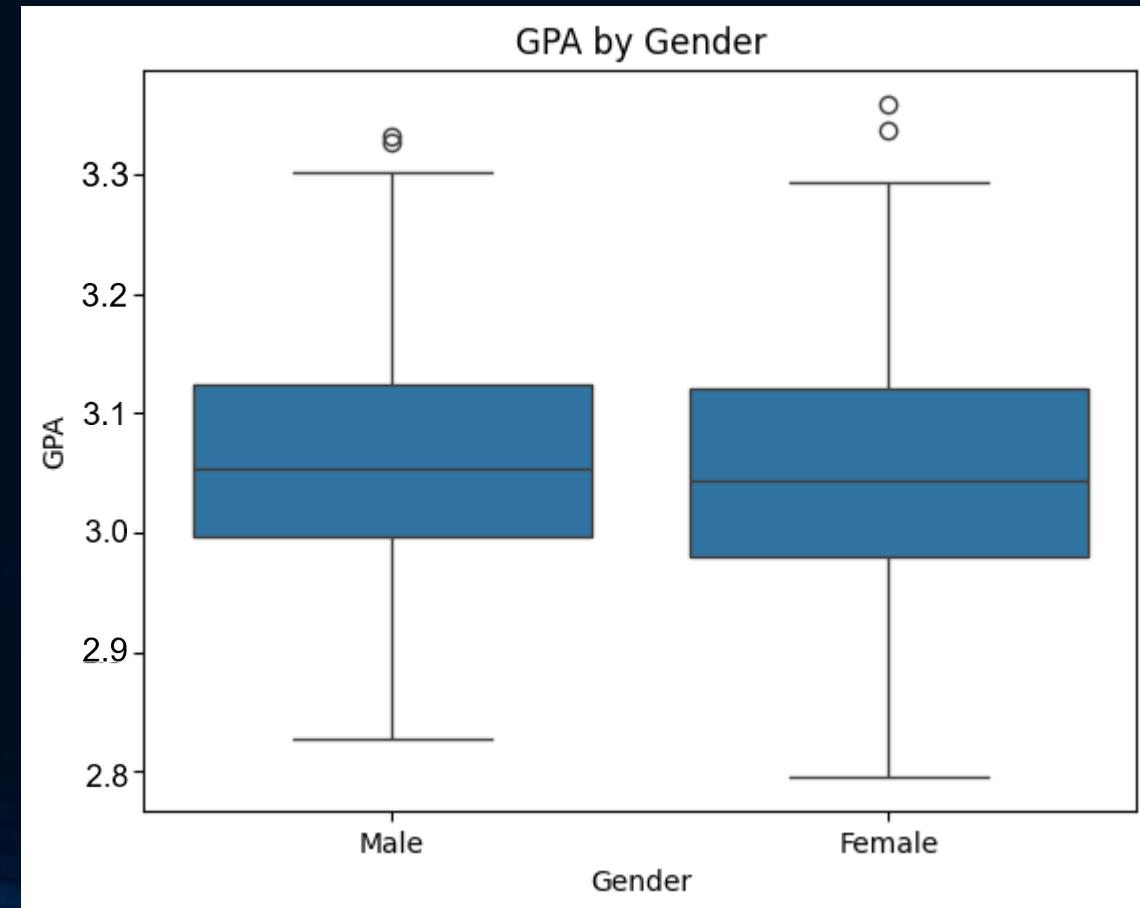
- Null Hypothesis (H_0): There is no correlation between GPA and attendance
- Alternative Hypothesis (H_1): There is significant correlation between GPA and attendance
- p-value: $0.0000 < 0.05 \rightarrow$ **Reject H_0**
- Conclusion: GPA increases with better attendance

Exploratory Analysis – Gender

Additional Exploration: GPA and Gender

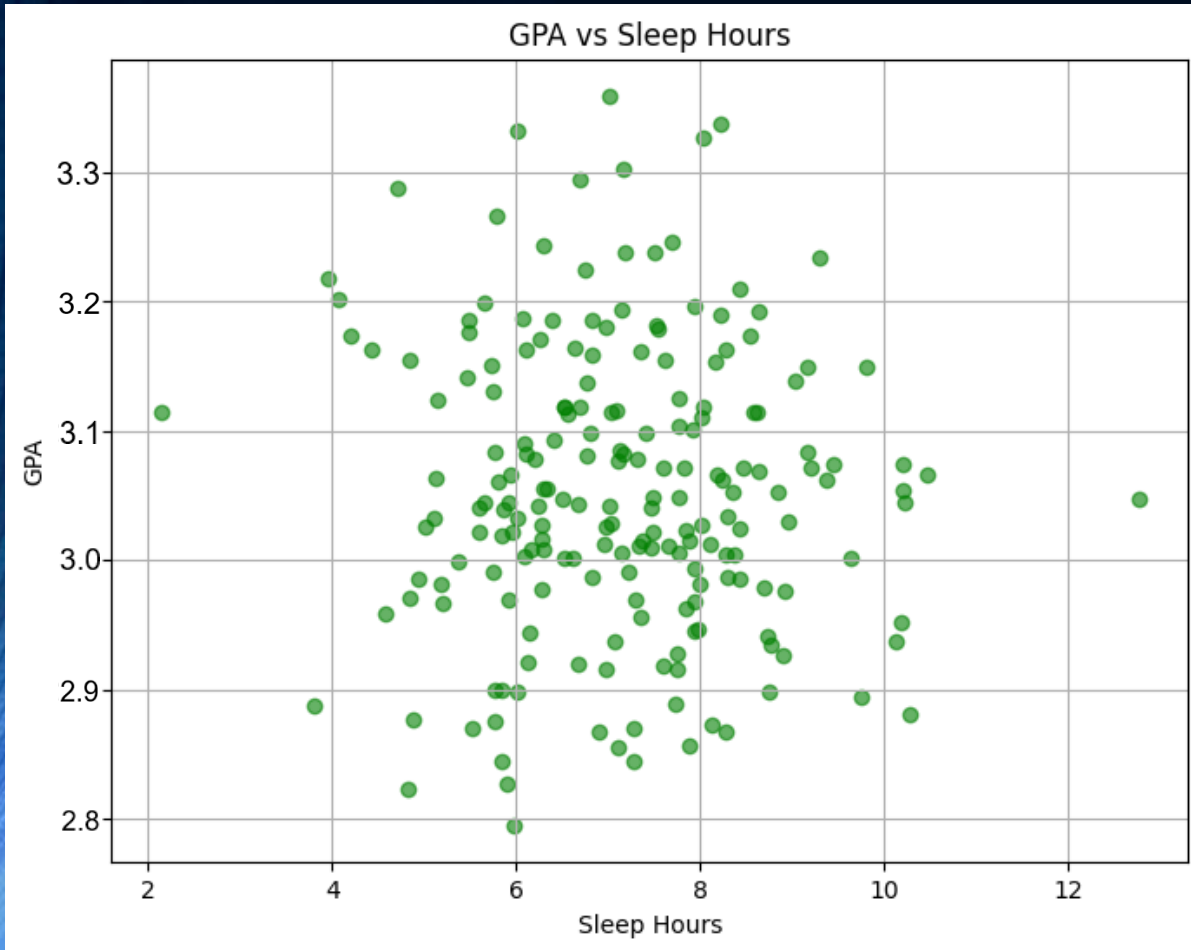
- Null Hypothesis (H_0): GPA_male = GPA_female
- P-value = 0.3659 > 0.05 → Fail to reject H_0 (**Not Significant**)
- No significant gender-based GPA difference

GPA by Gender (Box Plot)

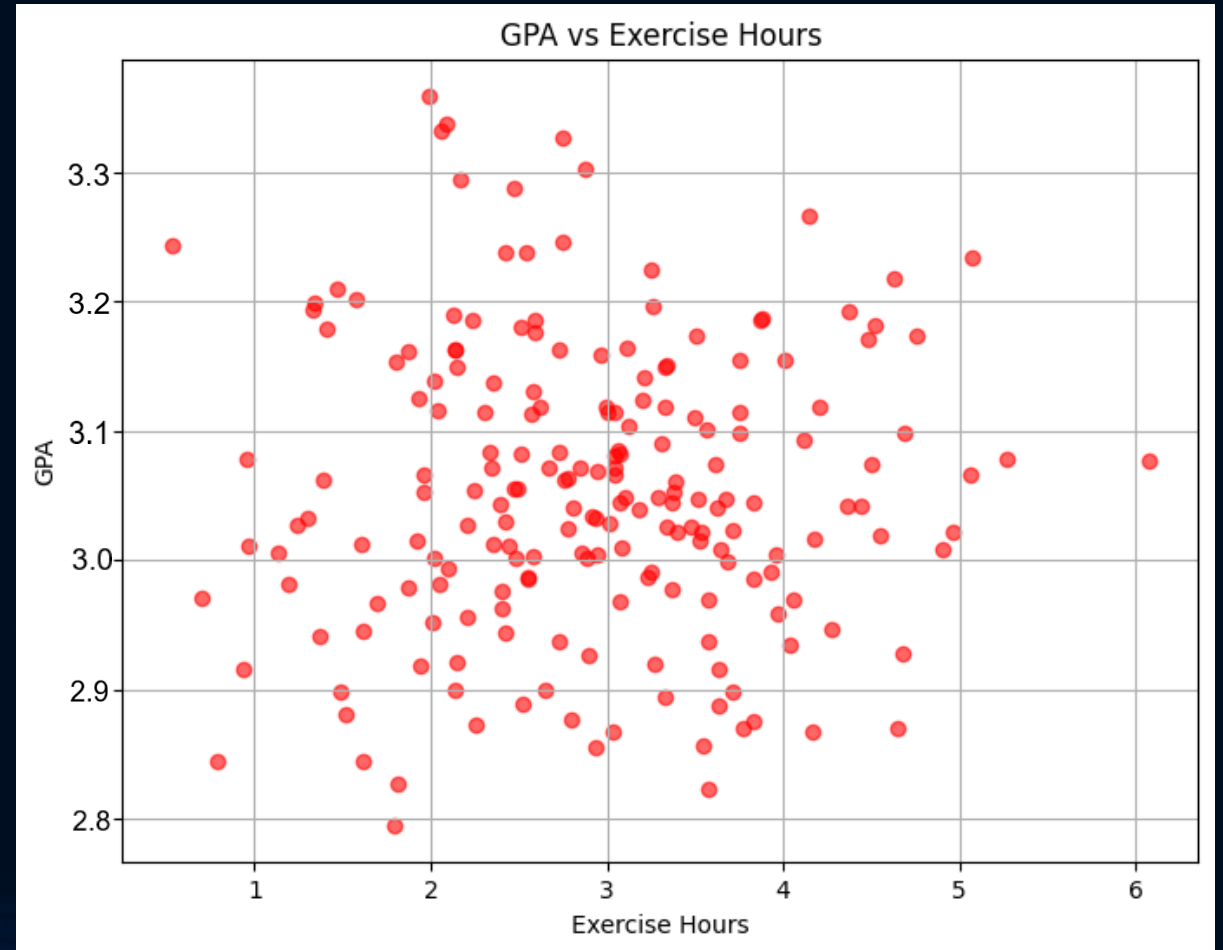


Visualizing Relationships

GPA vs Sleep Hours (Scatter Plot)

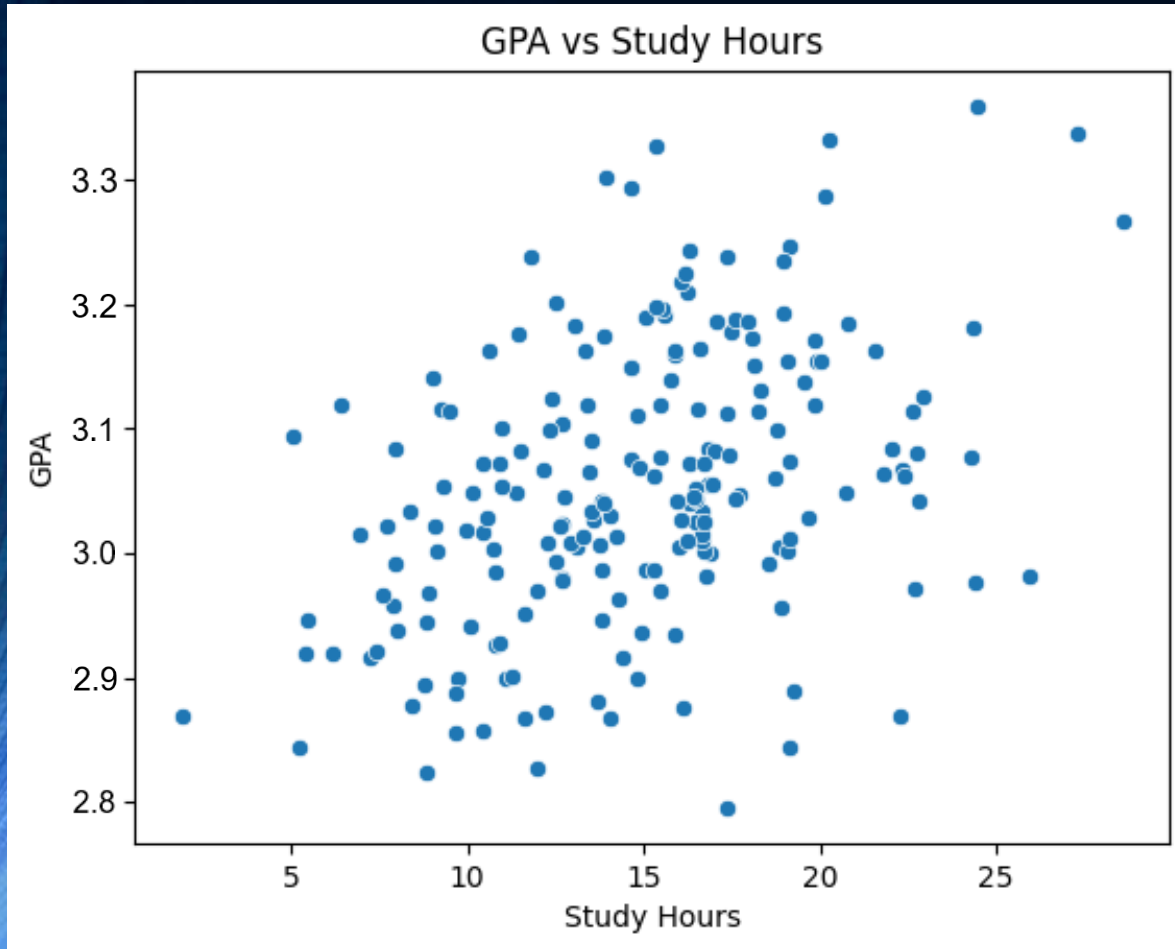


GPA vs Exercise Hours (Scatter Plot)

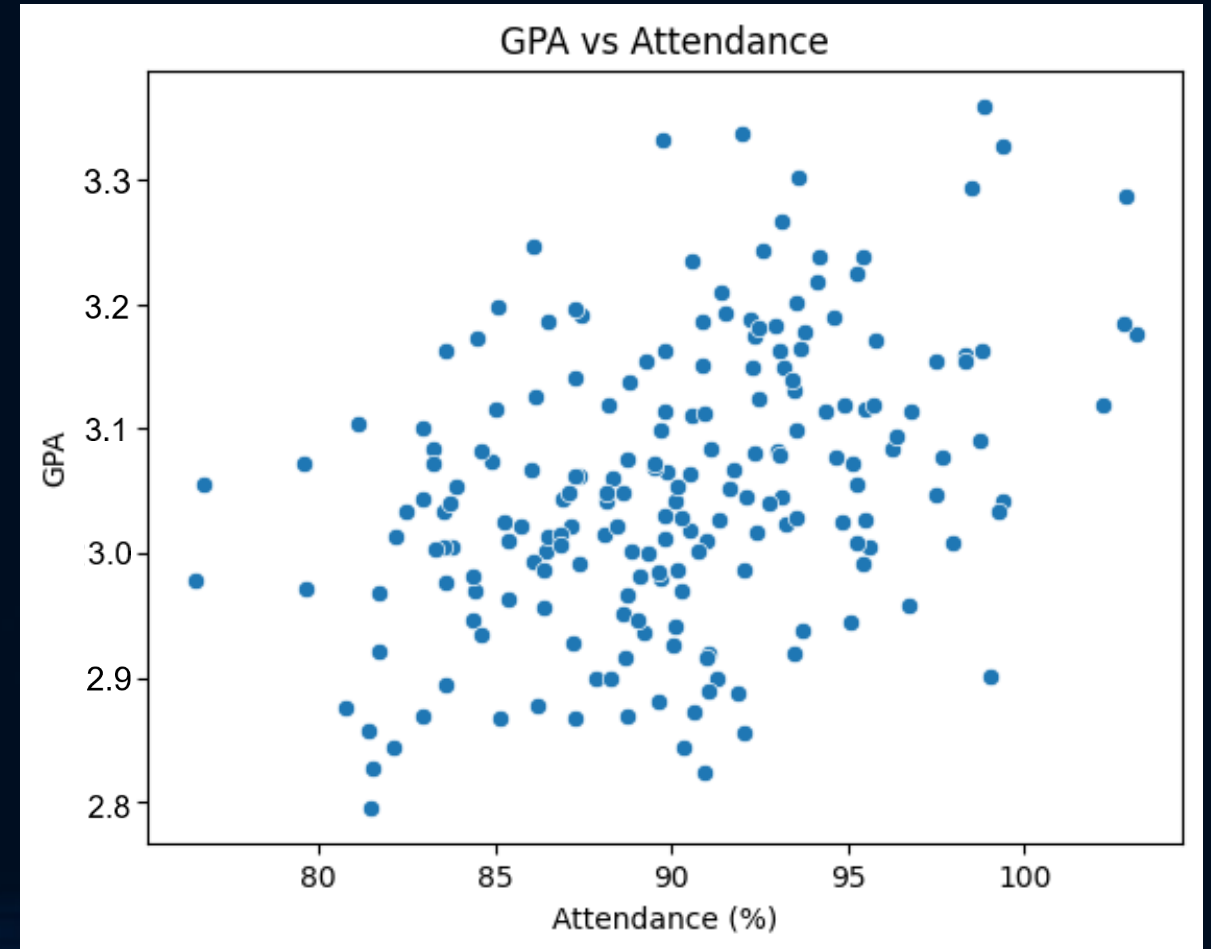


Visualizing Relationships

GPA vs Study Hours (Scatter Plot)



GPA vs Attendance (Scatter Plot)



Results

Key Takeaways

- **Important Predictors of GPA:**
→ **Study Hours** and **Attendance** significantly affect GPA. Regression shows study time is the best predictor of GPA.
- **Variables with Little to No Effect:**
→ **Sleep** and **Exercise** didn't show meaningful effects. No GPA gap between genders.
- **Practical Implication:**
→ To support academic success, encouraging consistent study habits and class attendance may be more effective than focusing on sleep or exercise (in this specific context).

Limitations and Future Work

- Synthetic data may miss real-world variance or noise
- Future: Also explore other factors like mental health (stress), social life, or job responsibilities (workload).

Thank You for your Attention